Interaction between fluids and Helfrich membranes

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1. [1] study the steady state with arc-length gauge
2. In the dynamics, arc-length gauge does not work
3. We show how to use a generalized arc-length gauge and how to correctly incorporate it into the dynamics to obtain a proper dynamics, **??**.
4. In this dynamics
   - we implement the Crank-Nicolson method to the equations written in the reference configuration, and prove it to be numerically stable
   - we show that the derivative of mesh motion $\dot{\boldsymbol{u}}_{\mathrm{D}}$ can be obtained exactly from an auxiliary PDE withotu recurring to error-prone time discretization.
5. We couple the membrane dynamics to a bulk fluid. The membrane is a complex elastic body -¿ We proceed by steps:
   (a) Bulk fluid + rigid body Fig. 2
   (b) Elastic body with stable elastic model (stable under compression)
   (c) Bulk fluid + elastic body with stable elastic model, Fig. 5
   (d) Bulk fluid + membrane, Fig. 7
      - Bulk fluid and membrane both described with Crank-Nicolson method -¿ we prove that the resulting dynamics is numerically stable. Say that you found a way to teplace $t \to t_n$ that is stable and that you could have used other ways which are also correct to within $\mathcal{O}(\Delta t)$.
      - It allows for overhangs
      - It is the first study that describes a bulk fluid coupled to a Helfrich membrane with ALE
      - It allows for turbulent behavior for both the bulk fluid and the membrane tangential flow

Perspectives:

1. Study nucleoid compaction in E. Coli with ALE
2. arc length gauge in 2d

I. FLUID AND RIGID BODY

In this Section, we will discuss the interaction between a bulk fluid (F) and the simplest structure—a rigid body (R) which is only allowed to turn about one point.

In all the systems that we will consider, we will introduce the reference (r) and current (c) configuration [2–4], see Fig. 1. All quantities relative to the r and c configuration will be denoted by the superscript r and c, respectively. In the r configuration, the region occupied by the bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\mathrm{r}}$. The axis of the ellipse, e..g, the rigid body, lies parallel to the $\mathsf{x}_{\mathrm{c}}^1$ axis. In the c configuration, the region occupied by the bulk fluid is described by Cartesian coordinates $\mathbf{x}_{\mathrm{c}}$. The axis of the e'llipse, e..g, the rigid body, forms an angle θ (red arc) with respect to the $\mathsf{x}_{\mathrm{c}}^1$ axis.

The r configuration, is related to the c one as follows [2], see Fig. 1. At time t, the point $\mathbf{x}_{\mathrm{c}}$ corresponds to a point $\mathbf{x}_{\mathrm{r}}$ in the reference configuration through the deformation field [5]

$$\boldsymbol{u}_{\mathrm{D}}^t(\mathbf{x}_{\mathrm{r}}) \equiv \mathbf{x}_{\mathrm{c}} - \mathbf{x}_{\mathrm{r}}. \tag{1}$$

and the deformation-gradient tensor

$$F_{\alpha\beta}(\boldsymbol{u}) \equiv \delta_{\alpha\beta} + \frac{\partial u^\alpha}{\partial \mathsf{x}_{\mathrm{r}}^\beta}, \tag{2}$$

where we set

$$\partial_\alpha^{\mathrm{r}} \equiv \frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^\alpha}, \tag{3}$$

$$\partial^{\mathrm{c}}_\alpha \equiv \frac{\partial}{\partial \mathsf{x}^\alpha_{\mathrm{c}}}. \quad (4)$$

Here, we will denote the mapping between $\mathbf{x}_{\mathrm{c}}$ and $\mathbf{x}_{\mathrm{r}}$ in Eq. (1) with the fields

$$\mathbf{x}_{\mathrm{c}} = \boldsymbol{\varphi}^t(\mathbf{x}_{\mathrm{r}}), \quad (5)$$
$$\mathbf{x}_{\mathrm{r}} = \boldsymbol{\psi}^t(\mathbf{x}_{\mathrm{c}}). \quad (6)$$

We denote the F velocity and surface tension by $v_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}})$ and $\sigma_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{c}})$, respectively, where the superscript c specifies that these fields depend on the coordinate $\mathbf{x}_{\mathrm{c}}$ in the c configuration.

The solution of the interaction dynamics between F and R is involved because of the presence of the moving boundary $\partial\Omega^{\mathrm{c}}_{\circ}$, cf. Fig. 1. Following the arbitrary Lagrangian-Eulerian (ALE) procedure, we will bridge between the Lagrangian and the Eulerian description used to describe, respectively, R and F [2]. To achieve this, in what follows we will write the equations of motion for R and F, which are naturally formulated the c configuration first, and then rewrite them in the r configuration.

A. Rigid body motion

In this Section, we will work out the equations of motion for R.

1. *Equation of motion in the current-configuration coordinates*

The dynamics of R is given by the equations of motion for a rigid body which, in c coordinates, read [6]

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = -\int_{\partial\Omega^{\mathrm{c}}_{\circ}} \mathrm{d}l_{\mathrm{c}}\, \epsilon_{3\alpha\beta}\, \sigma^{\mathrm{c}}_{\mathrm{F}\,\beta\gamma} \hat{n}^{\mathrm{c}\,\gamma}, \quad (7)$$
$$\omega \equiv \frac{\mathrm{d}\theta}{\mathrm{d}t}. \quad (8)$$

Equation (7) relates the angular acceleration of R to the torque of external forces. In what follows, greek indices $\alpha, \beta, \cdots$ will be used to denote vectors and tensors in two-dimensional Euclidean space—their position as upper or lower indices is thus immaterial [3, 7]. The moment of inertia of R is denoted by I. The right-hand side (RHS) of Eq. (7) contains the line integral along the boundary of R, of the torque of the local force exerted by F on R with respect to the left focal point $\boldsymbol{f}$ of R [5]. This force is expressed in terms of the F stress tensor [5]

$$\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta} \equiv \sigma_{\mathrm{F\,c}}\delta_{\alpha\beta} + \eta_{\mathrm{F}}\left(\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\alpha}}{\partial \mathsf{x}^\beta_{\mathrm{c}}} + \frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\beta}}{\partial \mathsf{x}^\alpha_{\mathrm{c}}}\right), \quad (9)$$

where $\epsilon_{\alpha\beta\gamma}$ is the three-dimensional Levi-Civita symbol, and η_{F} the two-dimensional viscosity of F. Finally, $\mathrm{d}l_{\mathrm{c}}$ is the line element of $\partial\Omega^{\mathrm{c}}_{\circ}$ in the c configuration, and $\hat{\boldsymbol{n}}^{\mathrm{c}}$ the unit boundary normal to $\partial\Omega^{\mathrm{c}}_{\circ}$ pointing outside Ω^{c}.

2. *Equation of motion in the reference-configuration coordinates*

$$I\frac{\mathrm{d}\omega}{\mathrm{d}t} = \int_{\partial\Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s\, \epsilon_{\alpha\beta}\left[\varphi^{\alpha,\,t}(\boldsymbol{\xi}(s)) - f^\alpha\right] \sigma^{\mathrm{r}}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s))\epsilon_{\gamma\lambda}F_{\lambda\mu}(\boldsymbol{u}^t_{\mathrm{D}}(\boldsymbol{\xi}(s)))\frac{\mathrm{d}\xi^\mu(s)}{\mathrm{d}s} \quad (10)$$

In Eq. (10), the integral in the RHS is over the curvilinear boundary $\partial\Omega^{\mathrm{r}}_{\circ}$, which is parametrized as $\mathbf{x}_{\mathrm{r}} = \boldsymbol{\xi}(s)$, where s is the curvilinear coordinate. Also, $\epsilon_{\alpha\beta}$ is the two-dimensional Levi-Civita symbol [7], and $\sigma^{\mathrm{r}}_{\mathrm{F}}$ denotes the stress tensor in the r configuration:

$$\sigma^{\mathrm{r}\,t}_{\mathrm{F}\,\alpha\beta}(\mathbf{x}_{\mathrm{r}}) \equiv \sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}(\boldsymbol{\varphi}^t(\mathbf{x}_{\mathrm{r}})) \quad (11)$$
$$= \sigma_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t)\delta_{\alpha\beta} + \eta_{\mathrm{F}}\Big[G_{\gamma\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}))\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{\alpha}}{\partial \mathsf{x}^\gamma_{\mathrm{r}}} +$$

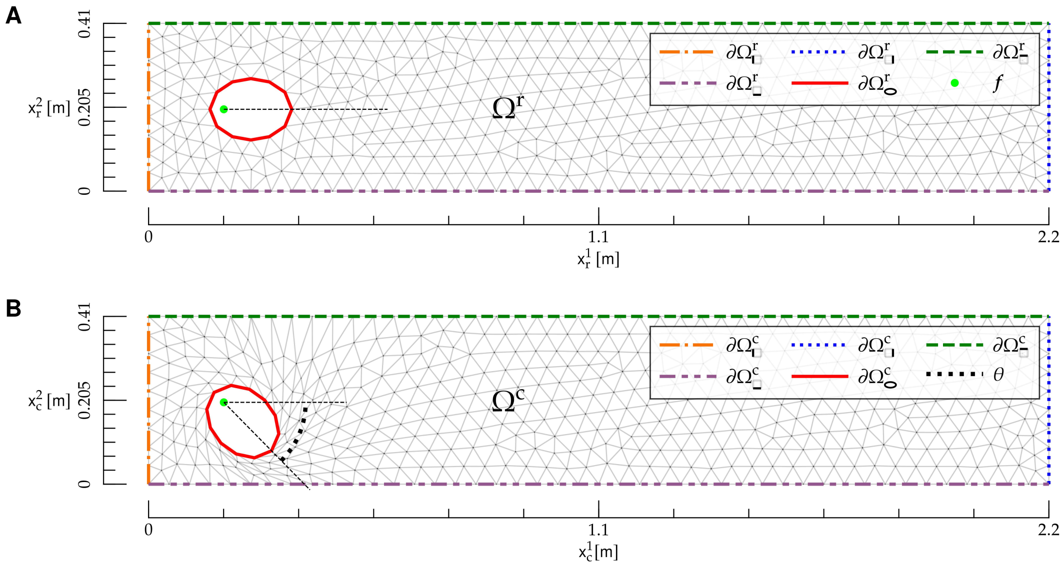

FIG. 1: Reference and current configuration for the interaction between a bulk fluid and a rigid body. **A**) Reference configuration. The region Ω^r is given by the rectangle $0 \le x_r^1 \le L$, $0 \le x_r^2 \le h$ with the elliptical hole, and is delimited by the mesh (gray lines). Boundaries in the reference configuration are denoted by $\partial\Omega^r_{\square}$, $\partial\Omega^r_{\blacksquare}$, $\partial\Omega^r_{=}$, $\partial\Omega^r_{\_}$ and $\partial\Omega^r_{\circ}$ (colored dashed lines). The ellipse focal point $\boldsymbol{f}$ (light green dot) is also shown. **B**) Current configuration. The region Ω^c is delimited by the mesh (gray lines). Boundaries in the current configuration are denoted by $\partial\Omega^c_{\square}$, $\partial\Omega^c_{\blacksquare}$, $\partial\Omega^c_{=}$, $\partial\Omega^c_{\_}$ and $\partial\Omega^c_{\circ}$ (colored dashed lines), and the rotational angle θ of the rigid body is shown as an arc (black dotted line). The Cartesian coordinates $\mathbf{x}_c$ are also shown.

$$G_{\gamma\alpha}(\boldsymbol{u}_D^t(\mathbf{x}_r))\frac{\partial v_{F_c}^{\beta}}{\partial x_r^{\gamma}}\Big],$$

where

$$G_{\alpha\beta}(\boldsymbol{u}) \equiv [F(\boldsymbol{u})]^{-1}_{\alpha\beta}. \tag{12}$$

B. Fluid motion

In this Section, we will work out the equations of motion for F.

1. *Equations of motion in the current-configuration coordinates*

The equations of motion, expressed in terms of the c fields v_{F_c} and σ_{F_c}, are

$$\rho_F\left(\partial_t v_{F_c}^{\alpha} + v_{F_c}^{\beta}\frac{\partial v_{F_c}^{\alpha}}{\partial x_c^{\beta}}\right) = \frac{\partial \sigma_{F_c}}{\partial x_c^{\alpha}} + \eta_F \frac{\partial}{\partial x_c^{\beta}}\frac{\partial v_{F_c}^{\alpha}}{\partial x_c^{\beta}}, \tag{13}$$

$$\frac{\partial v_{F_c}^{\alpha}}{\partial x_c^{\alpha}} = 0, \tag{14}$$

and they hold for $x_c \in \Omega^c$. Equations (13) and (14) are the Navier-Stokes (NS) and continuity equation, respectively, in a flat, two-dimensional geometry [5]—an example of the NS equations on curved geometry will be presented in X. There, ρ_F is the two-dimensional F density.

The boundary conditions (BCs) for Eqs. (7), (8), (13) and (14) are

$$v_{F_c}^{1}(\mathbf{x}_c) = \boldsymbol{v}_{F\square} \ \text{ on } \partial\Omega^c_{\square}, \tag{15}$$

$$v_{F_c}^{2}(\mathbf{x}_c) = 0 \ \text{ on } \partial\Omega^c_{\square}, \tag{16}$$

$$\boldsymbol{v}_{F_c}(\mathbf{x}_c) = \mathbf{0} \ \text{ on } \partial\Omega^c_{=}, \tag{17}$$

$$v_{F_c}^{\alpha}(\mathbf{x}_c) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathbf{x}_c) - f^{\beta}] \ \text{ on } \partial\Omega^c_{\circ}, \tag{18}$$

$$\eta_F \frac{\partial v_{F_c}^{\alpha}}{\partial x_c^1} = 0 \ \text{ on } \partial\Omega^c_{\blacksquare}, \tag{19}$$

$$\sigma_{F_c}(\mathbf{x}_c) = 0 \ \text{ on } \partial\Omega^c_{\blacksquare}. \tag{20}$$

Equations (15) and (16) ensure that F is injected on the boundary $\partial\Omega^c_{\square}$ in a direction parallel to the x_c^1 axis, and Eq. (17) is a no-slip BC [5] along the channel walls $\partial\Omega^c_{=}$, see Fig. 1. Equation (18) ensures that, at the boundary $\partial\Omega^c_{\circ}$, the F velocity matches the rotational velocity of R. Given that R can only rotate about its left focal point $\boldsymbol{f}$, the R rotational velocity of a point $\mathsf{x}_{\mathrm{c}} \in \partial\Omega^{\mathrm{c}}_{\circ}$ is given by

$$\frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}[\psi^{t\,\beta}(\mathsf{x}_{\mathrm{c}}) - c_\beta], \tag{21}$$

where $\boldsymbol{R}$ is the rotation matrix corresponding to a counterclockwise rotation:

$$\boldsymbol{R}(\theta) \equiv \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{22}$$

Finally, Eqs. (19) and (20) ensure, respectively, that on $\partial\Omega^{\mathrm{c}}_{\square}$ there is zero traction and tension; this corresponds to the hypothesis that there is free flow at $\partial\Omega^{\mathrm{c}}_{\square}$ [5, 8].

2. *Equations of motion in the reference-configuration coordinates*

For each field $\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathsf{x}_{\mathrm{c}}), \sigma_{\mathrm{F}}(\mathsf{x}_{\mathrm{c}})$ which depends on the coordinate x_{c} in the c configuration, we introduce its counterpart in the r configuration:

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathsf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\boldsymbol{\varphi}^t(\mathsf{x}_{\mathrm{r}}), t), \tag{23}$$

$$\sigma_{\mathrm{F}\,\mathrm{c}}(\mathsf{x}_{\mathrm{r}}, t) \equiv \boldsymbol{\sigma}_{\mathrm{F}\,\mathrm{r}}(\boldsymbol{\varphi}^t(\mathsf{x}_{\mathrm{r}}), t). \tag{24}$$

We will now rewrite the equations of motion and the BCs in terms of $\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}$, $\sigma_{\mathrm{F}\,\mathrm{c}}$, and the r coordinate x_{r}. The equations of motion (13) and (14) yield

$$\rho_{\mathrm{F}} \left\{ \frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\alpha}(\mathsf{x}_{\mathrm{r}}, t)}{\partial t} + \left[v_{\mathrm{F}\,\mathrm{r}}^{\gamma}(\mathsf{x}_{\mathrm{r}}, t) - \frac{\mathrm{d}\varphi^{\gamma\,t}(\mathsf{x}_{\mathrm{r}})}{\mathrm{d}t} \right] G_{\beta\gamma}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\alpha}(\mathsf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} \right\} = \tag{25}$$

$$G_{\beta\alpha}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial \sigma_{\mathrm{F}\,\mathrm{r}}(\mathsf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} + \eta_{\mathrm{F}}\, G_{\delta\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial}{\partial \mathsf{x}^{\delta}_{\mathrm{r}}} \left[G_{\gamma\beta}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial v_{\mathrm{F}\,\mathrm{r}}(\mathsf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\gamma}_{\mathrm{r}}} \right],$$

$$G_{\beta\alpha}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\alpha}(\mathsf{x}_{\mathrm{r}}, t)}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0, \tag{26}$$

which hold in Ω^{r}.

The BCs (15) to (20) are rewritten as

$$v_{\mathrm{F}\,\mathrm{r}}^{1}(\mathsf{x}_{\mathrm{r}}) = \boldsymbol{v}_{\mathrm{F}\,\square} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{27}$$

$$v_{\mathrm{F}\,\mathrm{r}}^{2}(\mathsf{x}_{\mathrm{r}}) = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{28}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}(\mathsf{x}_{\mathrm{r}}) = \boldsymbol{0} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{29}$$

$$v_{\mathrm{F}\,\mathrm{r}}^{\alpha}(\mathsf{x}_{\mathrm{r}}) = \frac{\mathrm{d}[R(\theta)]_{\alpha\beta}}{\mathrm{d}t}(\mathsf{x}^{\beta}_{\mathrm{r}} - f^{\beta}) \ \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{30}$$

$$\eta_{\mathrm{F}} G_{\beta 1}(\boldsymbol{u}^t_{\mathrm{D}}(\mathsf{x}_{\mathrm{r}})) \frac{\partial v_{\mathrm{F}\,\mathrm{r}}^{\alpha}}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{31}$$

$$\sigma_{\mathrm{F}\,\mathrm{r}}(\mathsf{x}_{\mathrm{r}}) = 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\square}. \tag{32}$$

C. Motion of bulk fluid domain

The equations of motion (10) and (25) to (32) involve the displacement field $\boldsymbol{u}_{\mathrm{D}}$, which describes how the region—or domain—which contains F, is deformed in time, cf. Fig. 1. We will denote this region as the D. We observe that this field has, so far, been left arbitrary, and we have the freedom to choose it.

Here, $\boldsymbol{u}_{\mathrm{D}}$ will be determined with an analogy with the theory of elasticity [2]. We imagine that, as R rotates about its focal point, each point $\mathsf{x}_{\mathrm{r}} \in \Omega^{\mathrm{r}}$ moves into a point $\mathsf{x}_{\mathrm{c}} \in \Omega^{\mathrm{c}}$, and that the domain Ω^{r} is deformed, e.g., locally stretched and compressed, into Ω^{c}, like an elastic medium.

We will thus determine $\boldsymbol{u}_{\mathrm{D}}$ as the solution of a boundary-value problem (BVP) given by the following elastic model [2, 9]

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\partial \mathsf{x}^{\beta}_{\mathrm{r}}} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{33}$$

where the elastic stress tensor S_{D} is given by

$$S_{\mathrm{D}\,\alpha\beta} \equiv F_{\alpha\gamma} T_{\gamma\beta}, \tag{34}$$

and

$$T_{\alpha\beta} \equiv K(\boldsymbol{u}_{\mathrm{D}}) E_{\gamma\gamma}\delta_{\alpha\beta} + 2\mu(\boldsymbol{u}_{\mathrm{D}}) \left(E_{\alpha\beta} - \frac{1}{2}\delta_{\alpha\beta}E_{\gamma\gamma} \right), \tag{35}$$

$$E_{\alpha\beta} \equiv \frac{1}{2}\left(F_{\gamma\alpha}F_{\gamma\beta} - \delta_{\alpha\beta} \right), \tag{36}$$

$$K(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}, \tag{37}$$

$$\mu(\boldsymbol{u}) \equiv \frac{1}{[\det(F(\boldsymbol{u}))]^{\zeta}}. \tag{38}$$

In Eqs. (37) and (38), we included an additional dependence of bulk modulus K and the shear modulus μ on the deformation-gradient tensor (2). In the absence of this dependence, the elastic model above would be unstable under local compression. In fact, the model's Lagrangian would not penalize configurations with small $\det F$. The inclusion of the dependence (37) and (38) on $\det F$ allows for penalizing these configurations—the larger the exponent $\zeta > 0$, the stronger the penalty. As a result, this dependence makes the model stable under mesh compression, such as the one shown below R in Fig. 1B.

The BCs for the partial differential equation (PDE) (33) are

$$\boldsymbol{u}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{39}$$
$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{f} - \mathbf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta)\cdot(\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}. \tag{40}$$

Equation (39) enforces the fact that the mesh is not deformed at the boundary $\partial\Omega^{\mathrm{r}}_{\square}$. Finally, Eq. (40) ensures that the mesh deformation at $\partial\Omega^{\mathrm{r}}_{\circ}$ matches the deformation induced by the rotation of R.

Equations (33), (39) and (40) constitute the BVP which determines $\boldsymbol{u}_{\mathrm{D}}$. Along the lines of BVPs in the theory of elasticity, such BVP is already, naturally formulated in the r configuration [3].

From the BVP above, we obtain a BVP for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, which will be needed in the following to describe the D motion. By deriving both sides of Eqs. (33), (39) and (40), we obtain, respectively,

$$\frac{\partial}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} \frac{\mathrm{d}S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{41}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{r}}_{\square}, \tag{42}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\cdot(\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{43}$$

where in Eqs. (42) and (43) we used the fact that $\mathbf{x}_{\mathrm{r}}$ is independent of time, and in Eq. (43) we substituted Eq. (8). The left-hand side (LHS) of Eq. (41) depends on the second partial derivatives of $\boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$—see Eqs. (34) to (38). As a result, given θ, ω, and $\boldsymbol{u}_{\mathrm{D}}$ from Eqs. (33), (39) and (40), the BVP (41) to (43) yields $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

Overall Eqs. (8), (10), (25) to (33) and (39) to (43) constitute a BVP for $\theta, \omega, \boldsymbol{v}_{\mathrm{F\,r}}, \sigma_{\mathrm{F\,r}}, \boldsymbol{u}_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}$ which, combined with the temporal BCs

$$\theta(t=0) = \theta_0 \tag{44}$$
$$\omega(t=0) = \omega_0 \tag{45}$$
$$\boldsymbol{v}_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t=0) = \boldsymbol{v}_{\mathrm{F\,r}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{46}$$
$$\boldsymbol{\sigma}_{\mathrm{F\,r}}(\mathbf{x}_{\mathrm{r}}, t=0) = \boldsymbol{\sigma}_{\mathrm{F\,r}\,0}(\mathbf{x}_{\mathrm{r}}), \tag{47}$$

determine the dynamics of the system.

D. Temporal discretization

To numerically solve for the dynamics in a time span $0 \le t \le T$, we discretize time by setting $\Delta t \equiv T/N$, $t_n \equiv n\,\Delta t$, with $n = 0, 1, \cdots, N$.

We combine the Crank Nicolson (CN) discretization method for NS equations [8, 10] with the ALE formulation, by discretizing the dynamical equations for R, F and D—see below.

1. *Rigid body*

Proceeding along the CN scheme [10], we define fields at staggered, integer and semi-integer time steps scheme [8]. We set

$$\sigma_{\mathrm{F\,r}}{}^{n}(\mathbf{x}_{\mathrm{r}}) \equiv \sigma_{\mathrm{F\,c}}(\mathbf{x}_{\mathrm{r}}, t_n) \tag{48}$$
$$\sigma_{\mathrm{F\,r}}{}^{n-1/2}(\mathbf{x}_{\mathrm{r}}) \equiv \sigma_{\mathrm{F\,c}}\left(\mathbf{x}_{\mathrm{r}}, \frac{t_n + t_{n-1}}{2}\right), \tag{49}$$

and similarly for other quantities.

We discretize Eqs. (8) and (10) and, neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\frac{\theta^n - \theta^{n-1}}{\Delta t} = \omega^n, \tag{50}$$
$$\frac{\omega^n - \omega^{n-1}}{\Delta t} = I \int_{\partial\Omega^{\mathrm{r}}_{\circ}} \mathrm{d}s\, \epsilon_{\alpha\beta} \left[\xi^{\alpha}(s) + u^{n-1,\,\alpha}(\boldsymbol{\xi}(s)) - f^{\alpha}\right] \sigma^{\mathrm{r}\;n-1}_{\mathrm{F}\,\beta\gamma}(\boldsymbol{\xi}(s); \boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \sigma_{\mathrm{F\,r}}{}^{n-3/2})\times \tag{51}$$

$$\epsilon_{\gamma\lambda} F_{\lambda\mu}(\boldsymbol{u}_{\mathrm{D}}^{n-1}(\boldsymbol{\xi}(s)))\,\frac{\mathrm{d}\xi^{\mu}(s)}{\mathrm{d}s},$$

where in Eq. (51) we used Eqs. (1) and (5) and we explicitly indicated the dependence of the stress tensor (11) on $\boldsymbol{v}_{\mathrm{F_c}}$ and $\sigma_{\mathrm{F_c}}$.

2. *Bulk fluid*

To discretize in time the F problem, we will apply the CN splitting scheme [8, 10] to the NS equations (25) and (26) in the r configurations. This splitting scheme is derived form the incremental pressure correction scheme [11], in which the solution of the NS equations is split into three intermediate steps. In what follows, we will synonymously use the terms 'pressure' and 'surface tension' for conciseness [12].

The discrete form of the PDEs Eqs. (25) and (26) is

$$\rho_{\mathrm{F}}\left[\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} - {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\Delta t} + \frac{3}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\gamma} - \dot{u}^{n-1,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} - \frac{1}{2}({v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\gamma} - \dot{u}^{n-2,\,\gamma})\times G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n-2,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right] = G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial {\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} + \eta_{\mathrm{F}}\,G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\delta}}\left[G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} + {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2}\right)\right], \tag{52}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial {v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}} = 0, \tag{53}$$

where we omitted the dependence on $\mathbf{x}_{\mathrm{r}}$ for clarity. Also, in the LHS of Eq. (52) we rewrote the convective term according to the Adams-Bashforth discretization scheme [8, 13], and we used Eqs. (1) and (5) to rewrite the time derivative of $\boldsymbol{\varphi}$ in terms of $\dot{\boldsymbol{u}}_{\mathrm{D}}$.

The discrete form of the BCs (27) to (32) reads

$$\begin{aligned}
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,1} &= \boldsymbol{v}_{\mathrm{F}\,\sqsubset} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, && (54)\\
{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,2} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsubset}, && (55)\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n} &= \boldsymbol{0} \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqcup}, && (56)\\
{\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}} &= \omega^{n}\,\left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_n} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega^{\mathrm{r}}_{\circ}, && (57)\\
\eta_{\mathrm{F}}\,G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\left(\frac{{v_{\mathrm{F}}}_{\mathrm{r}}^{n,\,\alpha} + {v_{\mathrm{F}}}_{\mathrm{r}}^{n-1,\,\alpha}}{2}\right) &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}, && (58)\\
{\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2} &= 0 \ \text{ on } \partial\Omega^{\mathrm{r}}_{\sqsupset}. && (59)
\end{aligned}$$

We will now split the BVP into three steps [8, 14]:

1. **Approximated velocity.** We introduce

the velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$, which approximates the true velocity $\boldsymbol{v}_{\mathrm{F\,r}}$, and which satisfies the BVP

$$\rho_{\mathrm{F}}\left\{\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F\,r}}}^{n-1,\,\alpha}}{\Delta t}+\left[\frac{3}{2}({v_{\mathrm{F\,r}}}^{n-1,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-1,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-1})-\frac{1}{2}({v_{\mathrm{F\,r}}}^{n-2,\,\gamma}-\dot{u}_{\mathrm{D}}^{n-2,\,\gamma})\,G_{\beta\gamma}(\boldsymbol{u}_{\mathrm{D}}^{n-2})\right]\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right\}= \tag{60}$$

$$G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial {\sigma_{\mathrm{F\,r}}}^{*}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}+\eta_{\mathrm{F}}\,G_{\delta\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\delta}}\left(G_{\gamma\beta}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\right),$$

$$\overline{v}_{\mathrm{F}}^{1}=\boldsymbol{v}_{\mathrm{F}\,|\square}\ \text{ on } \partial\Omega^{\mathrm{r}}_{|\square}, \tag{61}$$

$$\overline{v}_{\mathrm{F}}^{2}=0\ \text{ on } \partial\Omega^{\mathrm{r}}_{|\square}, \tag{62}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\boldsymbol{0}\ \text{ on } \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{63}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}=\omega^{n}\,\left.\frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta}\right|_{\theta=\theta_{n}}\cdot(\mathbf{x}_{\mathrm{r}}-\boldsymbol{f})\ \text{on } \partial\Omega^{\mathrm{r}}_{\circ}, \tag{64}$$

$$\eta_{\mathrm{F}}\,G_{\beta 1}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial V_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}=0\ \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}. \tag{65}$$

where

$$\boldsymbol{V}_{\mathrm{F}}\equiv\frac{\overline{\boldsymbol{v}}_{\mathrm{F}}+{\boldsymbol{v}_{\mathrm{F\,r}}}^{n-1}}{2}, \tag{66}$$

$${\sigma_{\mathrm{F\,r}}}^{*}\equiv{\sigma_{\mathrm{F\,r}}}^{n-3/2}. \tag{67}$$

2. **Pressure correction.**

   Subtracting Eqs. (52) and (60) and neglecting $\mathcal{O}(\Delta t)$, we obtain

$$\rho_{\mathrm{F}}\frac{\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F\,r}}}^{n,\,\alpha}}{\Delta t}=G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\ \text{ in } \Omega^{\mathrm{r}}, \tag{68}$$

   where

$$\phi_{\mathrm{F}}\equiv{\sigma_{\mathrm{F\,c}}}^{*}-{\sigma_{\mathrm{F\,c}}}^{n-1/2} \tag{69}$$

   is the surface-tension increment. By applying $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\partial/\partial \mathsf{x}_{\mathrm{r}}^{\gamma}$ to both sides of Eq. (68) and using Eq. (53), we obtain a second-order PDE for ϕ_{F}:

$$\frac{\rho_{\mathrm{F}}}{\Delta t}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial\overline{v}_{\mathrm{F}}^{\alpha}}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}= \tag{70}$$

$$G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial}{\partial \mathsf{x}_{\mathrm{r}}^{\gamma}}\left(G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}\right).$$

   We will now work out the BCs for Eq. (70). Equations (59), (67) and (69) imply the BC

$$\phi_{\mathrm{F}}=0\ \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}. \tag{71}$$

   Multiplying Eq. (68) by $G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}$, where $\hat{\boldsymbol{n}}^{\mathrm{r}}$ is the unit boundary normal in the r configuration, we obtain

$$\frac{\rho_{\mathrm{F}}}{\Delta t}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\hat{n}^{\mathrm{r}\,\gamma}(\overline{v}_{\mathrm{F}}^{\alpha}-{v_{\mathrm{F\,r}}}^{n,\,\alpha})= \tag{72}$$

$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}.$$

   Combining Eq. (72) with Eqs. (54) to (57) and (61) to (64), we obtain the second BC

$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\,\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}= \tag{73}$$

$$0\ \text{on } \partial\Omega^{\mathrm{r}}_{|\square}\cup\partial\Omega^{\mathrm{r}}_{\underline{\square}}\cup\partial\Omega^{\mathrm{r}}_{\circ}.$$

   The BVP (70), (71) and (73) determines ϕ_{F} and, through the definition (69), the surface tension ${\sigma_{\mathrm{c}}}^{n-1/2}$.

3. **Velocity correction.** The velocity ${\boldsymbol{v}_{\mathrm{F\,r}}}^{n}$ is determined from the approximate velocity $\overline{\boldsymbol{v}}_{\mathrm{F}}$ from Eq. (68).

3. *Bulk fluid domain*

The discrete version of the BVP for $\boldsymbol{u}_{\mathrm{D}}$, Eqs. (33), (39) and (40), reads

$$\frac{\partial S_{\mathrm{D}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{D}}^{n})}{\partial \mathsf{x}_{\mathrm{r}}^{\beta}}=0\ \text{ in } \Omega^{\mathrm{r}}, \tag{74}$$

$$\boldsymbol{u}_{\mathrm{D}}^{n} = \mathbf{0} \text{ on } \partial\Omega_{\square}^{\mathrm{r}}, \tag{75}$$

$$\boldsymbol{u}_{\mathrm{D}}^{n} = \boldsymbol{f} - \mathbf{x}_{\mathrm{r}} + \boldsymbol{R}(\theta^n) \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega_{\circ}^{\mathrm{r}}, \tag{76}$$

and that of the BVP for $\dot{\boldsymbol{u}}_{\mathrm{D}}$, Eqs. (41) to (43), is

$$\frac{\partial}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}} \frac{\mathrm{d}S_{\mathrm{D}\,\alpha\beta}(\dot{\boldsymbol{u}}_{\mathrm{D}}^{n})}{\mathrm{d}t} = 0 \text{ in } \Omega^{\mathrm{r}}, \tag{77}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}}^{n} = \mathbf{0} \text{ on } \partial\Omega_{\square}^{\mathrm{r}}, \tag{78}$$

$$\dot{\boldsymbol{u}}_{\mathrm{D}}^{n} = \omega \frac{\mathrm{d}\boldsymbol{R}}{\mathrm{d}\theta} \cdot (\mathbf{x}_{\mathrm{r}} - \boldsymbol{f}) \text{ on } \partial\Omega_{\circ}^{\mathrm{r}}, \tag{79}$$

We can now iterate in time for the ensemble of fields as follows. At each step n, given θ^{n-1}, ω^{n-1}, ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}$, ${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-3/2}$, $\boldsymbol{u}_{\mathrm{D}}^{n-1}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n-1}$ from the preceding step, we

1. **Update R.** Solve the algebraic equations Eqs. (50) and (51) for θ^n and ω^n.
2. **Update D.** Solve the BVPs (74) to (79) for $\boldsymbol{u}_{\mathrm{D}}^{n}$ and $\dot{\boldsymbol{u}}_{\mathrm{D}}^{n}$.
3. **Update F.** Solve the BVPs (60) to (65) and (71) to (73) for $\overline{\boldsymbol{v}}_{\mathrm{F}}$ and ${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2}$, and obtain ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n}$ from Eq. (68).

The fields ${\boldsymbol{v}_{\mathrm{F}}}_{\mathrm{r}}^{n-1}$, ${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-3/2}$ in the c configuration are then obtained from the respective fields in the r configuration through Eqs. (23) and (24).

An example of such dynamics is shown in Fig. 2, where we study the interaction between air (F) and an aerogel (R)—an ultra light solid material involving a large porous structure [15]. The channel dimension are $L = 2.2\,\mathrm{m}$, $h = 0.41\,\mathrm{m}$, which have been taken from the Finite Element Analysis Tools 2D (FEAT2D) DFG 2D-3 benchmark example for a fuid flow around a cylinder [16, 17], see Fig. 1. Given that the materials, such as F and B, are two dimensional, in what follows we will estimate their parameter values by considering their three-dimensional counterparts, and assume a unit thickness in the out-of-plane direction. The parameters for B have been estimated from the properties of air: $\rho_{\mathrm{F}} = 1\,\mathrm{Kg/m^2}$, $\eta_{\mathrm{F}} = 10^{-5}\,\mathrm{Pa\,m\,sec}$ [18]. The inflow velocity profile is chosen as a Poiseuille flow

$$\boldsymbol{v}_{\mathrm{F}\,\square}(\mathbf{x}_{\mathrm{c}}) = \frac{6v_0}{h^2}\mathrm{x}_{\mathrm{c}}^2(\mathrm{x}_{\mathrm{c}}^2 - h), \tag{80}$$

with $v_0 = 1\,\mathrm{m/\,sec}$. The D mesh stiffness exponent has been chosen to be $\zeta = 3$ [2]. Finally, R is an ellipse with semi-axes $a = 0.1\,\mathrm{m}$, $b = 0.075\,\mathrm{m}$, center located at $\mathbf{x}_{\mathrm{r}} = (0.25\,\mathrm{m}, 0.2\,\mathrm{m})$, and moment of inertia $I = 10^{-3}\,\mathrm{Kg\,m^2}$, which corresponds to an aerogel density of $\sim 6\,\mathrm{Kg/m^3}$ [19].

As shown in Fig. 3, the dynamics of Fig. 2 involves a characteristic time scales τ_{B}, i.e., the period of the angular oscillations of B. This time scale can be estimated by considering the dynamical equation (7) for B, and approximating its LHS as I/τ_{B}^2 and its RHS as $(\eta_{\mathrm{F}} v_0/a\, 2\pi a)a$. Here, $\eta_{\mathrm{F}} v_0/a$ is the viscous component of the force per unit length exerted by F on B, cf. Eq. (9), $2\pi a$ estimates the total length of the boundary of B, and a approximates the lever arm of the force above. Putting everything together, we obtain

$$\tau_{\mathrm{B}} \sim \sqrt{\frac{I}{2\pi a \eta_{\mathrm{F}} v_0}}. \tag{81}$$

For the parameters of Fig. 2, Eq. (81) yields $\tau_{\mathrm{B}} \sim 130$ sec, which is in agreement with the period of the oscillations in Fig. 3.

II. FLUID AND ELASTIC BODY

In this Section, we will build on the analysis of Section I and put F into interaction with a more complex physical object—an elastic body (E). Only the main results will be presented here; their derivation follows the lines of Section I.

The r and c configurations are shown in Fig. 4. Here, the E boundary $\partial\Omega_{\bigcirc}$ may deform, but the inner E boundary $\partial\Omega_{\odot}$ is pinned to the $\mathrm{x}_{\mathrm{c}}^1$, $\mathrm{x}_{\mathrm{c}}^2$ plane.

The equations of motion are the following.

First, the dynamics of F is governed by the NS and mass-conservation equations (13) and (14), respectively. Their BCs are Eqs. (15) to (17), (19) and (20), which stay unchanged. On the other hand, BC (18), which ensures that the F velocity matches the B velocity at the interface between F and B, now reads

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathbf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{E}}(\mathbf{x}_{\mathrm{r}})}{\mathrm{d}t}\right|_{\mathbf{x}_{\mathrm{r}} = \psi^t(\mathbf{x}_{\mathrm{c}})} \text{ on } \partial\Omega_{\bigcirc}^{\mathrm{c}} \tag{82}$$

Second, the motion of E is governed by Newton's equations of motion for an elastic body [3]

$$\rho_{\mathrm{E}} \frac{\mathrm{d}^2 u_{\mathrm{E}}^{\alpha}(\mathbf{x}_{\mathrm{r}})}{\mathrm{d}t^2} = \frac{\partial S_{\mathrm{E}\,\alpha\beta}(\boldsymbol{u}_{\mathrm{E}})}{\partial \mathrm{x}_{\mathrm{r}}^{\beta}}, \tag{83}$$

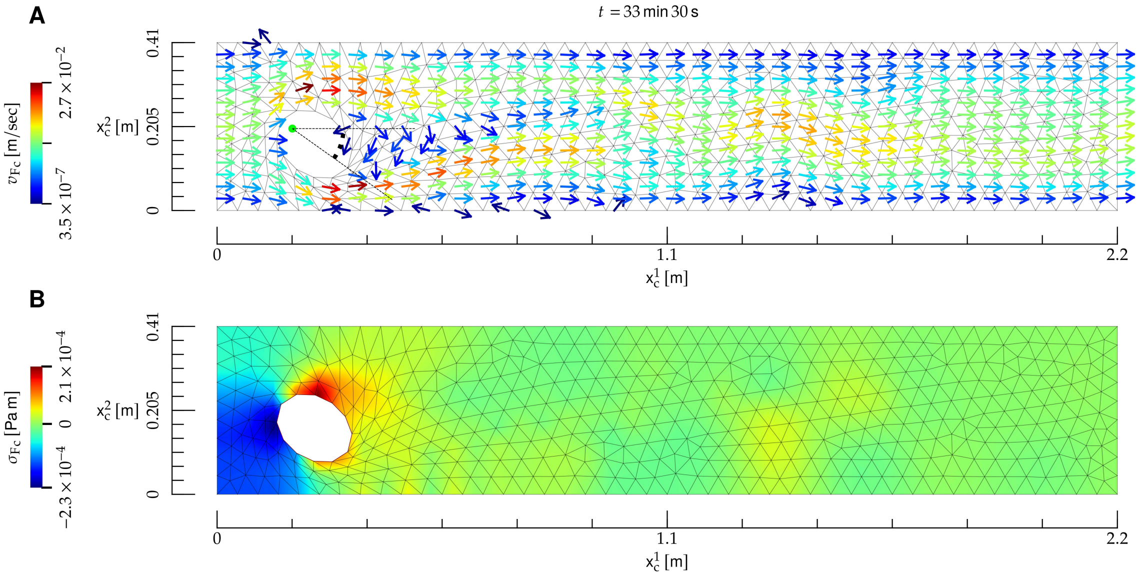

FIG. 2: Interaction between a bulk fluid (F) and a rigid rigid body (R), whose material properties roughly correspond to air (F) and an aerogel (R). **A**) Temporal snapshot of the mesh (black lines) and velocity field (arrows) at the current time t, shown on top. The direction of the velocity field is indicated by the arrows, and its norm by their color; the direction of arrows with velocity close to zero is not defined. Here, R (ellipse) is allowed to pivot about its left focal point (red dot), and the related angle with respect to the $\mathrm{x}_{\mathrm{c}}^{1}$ direction is denoted by a red arc. Here, the Reynolds number is $R \sim 2 \times 10^2$. **B**) Color map of the surface tension at the current instant of time t.

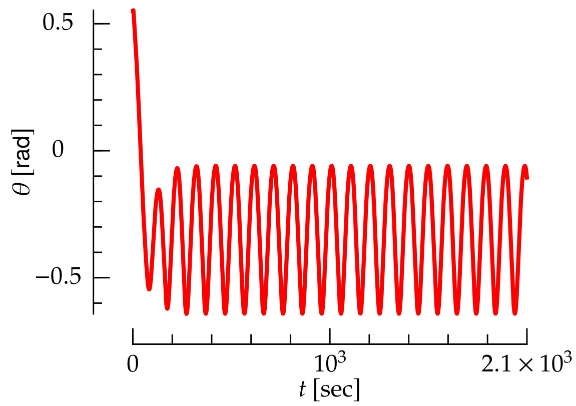

FIG. 3: Angle of the rigid body as a function of time for the interaction between a bulk fluid and a rigid body shown in Fig. 2.

where ρ_{E} is the density of E which is assumed to be independent of space. Also, $\boldsymbol{u}_{\mathrm{E}}$ is the deformation field of E, which is defined along the lines of Eq. (1). The E stress tensor is derived from a compressible neo-Hookean model [20], and it reads

$$S_{\mathrm{E}\alpha\beta} \equiv \frac{\mu}{\det(F)}\left(-\frac{1}{2}C_{\gamma\gamma}G_{\alpha\beta} + F_{\beta\alpha}\right) + K[(\det(F))^2 - 1]G_{\alpha\beta}, \tag{84}$$

where

$$C_{\alpha\beta} \equiv \delta_{\alpha\beta} + 2E_{\alpha\beta}. \tag{85}$$

We have chosen this model because it is stable under compression, i.e., its potential energy diverges as $\det(F) \to 0$ [20]. As opposed to linear models [3] and to the elastic model of Section I C, this model proves to yield a stable dynamics for E as it is compressed by F, see below and Fig. 5. The BCs for Eq. (83) are

$$\boldsymbol{u}_{\mathrm{E}}(\mathbf{x}_{\mathrm{r}}) = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\odot}, \tag{86}$$

$$S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}^t_{\mathrm{E}})\big|_{\mathbf{x}_{\mathrm{r}}=\boldsymbol{\xi}(s)}\,\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s} = \sigma^{\mathrm{c}}_{\mathrm{F}\alpha\beta}(\boldsymbol{\varphi}^t(\boldsymbol{\xi}(s)))\epsilon_{\beta\gamma}\frac{\mathrm{d}\varphi^{t,\gamma}(\boldsymbol{\xi}(s))}{\mathrm{d}s} \text{ on } \partial\Omega^{\mathrm{r}}_{\boldsymbol{\bigcirc}}. \tag{87}$$

In the LHS of Eq. (87), the quantity $\epsilon_{\beta\gamma}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s}$ represents the unit normal to $\partial\Omega^{\mathrm{r}}_{\boldsymbol{\bigcirc}}$, and similarly for the RHS.

Finally, the equations of motion for D are (33) and (41), with BCs (39) and (42) and

$$\boldsymbol{u}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}) = \boldsymbol{u}^t_{\mathrm{E}}(\mathbf{x}_{\mathrm{r}}) \text{ on } \partial\Omega^{\mathrm{r}}_{\boldsymbol{\bigcirc}}, \tag{88}$$

$$\dot{\boldsymbol{u}}^t_{\mathrm{D}}(\mathbf{x}_{\mathrm{r}}) = \dot{\boldsymbol{u}}^t_{\mathrm{E}}(\mathbf{x}_{\mathrm{r}}) \text{ on } \partial\Omega^{\mathrm{r}}_{\boldsymbol{\bigcirc}}. \tag{89}$$

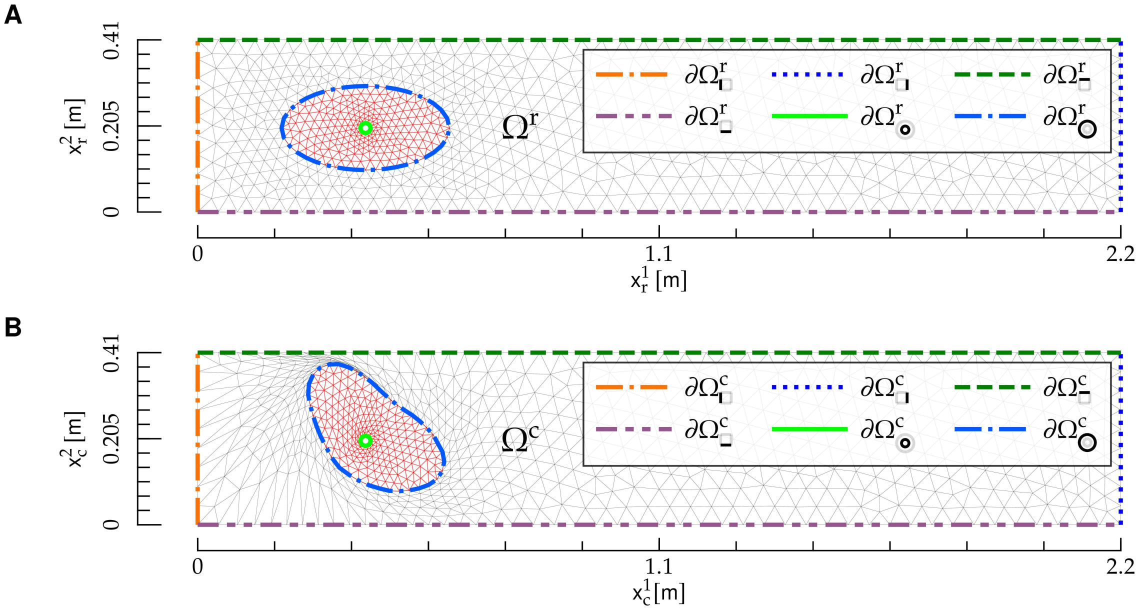

FIG. 4: Reference and current configuration for the interaction between a bulk fluid and an elastic body. The figures follows the same notation as Fig. 1.

Equation (88) and Eq. (89) ensure that the displacement and velocity of E and D are conforming at the E-D interface, and they replace Eqs. (40) and (43), respectively.

The system dynamics is obtained as follows: given ${\boldsymbol{v}_{\mathrm{F}_\mathrm{r}}}^{n-1}$, ${\sigma_{\mathrm{F}_\mathrm{r}}}^{n-3/2}$, $\boldsymbol{u}_\mathrm{E}^{n-1}$, $\dot{\boldsymbol{u}}_\mathrm{E}^{n-1}$, $\dot{\boldsymbol{u}}_\mathrm{D}^{n-1}$ and $\dot{\boldsymbol{u}}_\mathrm{D}^{n-1}$ from the preceding step, we

- **Update E.** Solve for $\boldsymbol{u}_\mathrm{E}^n$ and $\dot{\boldsymbol{u}}_\mathrm{E}^n$ the discrete version of Eq. (83)

$$\rho_\mathrm{E}\frac{\dot{u}_\mathrm{E}^{n,\,\alpha}-\dot{u}_\mathrm{E}^{n-1,\,\alpha}}{\Delta t}=\frac{\partial S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}_\mathrm{E}^n)}{\partial \mathrm{x}_\mathrm{r}^\beta}, \tag{90}$$

$$\frac{\boldsymbol{u}_\mathrm{E}^n-\boldsymbol{u}_\mathrm{E}^{n-1}}{\Delta t}=\dot{\boldsymbol{u}}_\mathrm{E}, \tag{91}$$

with BCs obtained from Eqs. (86) and (87):

$$\boldsymbol{u}_\mathrm{E}^n=0 \text{ on } \partial\Omega^\mathrm{r}_{\odot}, \tag{92}$$

$$\epsilon_{\beta\gamma}\; S_{\mathrm{E}\alpha\beta}(\boldsymbol{u}_\mathrm{E}^n)\big|_{\mathrm{x}_\mathrm{r}=\boldsymbol{\xi}(s)}\frac{\mathrm{d}\xi^\gamma}{\mathrm{d}s}= \tag{93}$$

$$\sigma^\mathrm{r}_{\mathrm{F}\alpha\beta}(\boldsymbol{\xi}(s);{\boldsymbol{v}_{\mathrm{F}_\mathrm{r}}}^{n-1},{\sigma_{\mathrm{F}_\mathrm{r}}}^{n-3/2},\boldsymbol{u}_\mathrm{E}^n)\epsilon_{\beta\gamma}\; F_{\gamma\delta}(\boldsymbol{u}_\mathrm{E}^{n-1})\big|_{\mathrm{x}_\mathrm{r}=\boldsymbol{\xi}(s)}\frac{\mathrm{d}\xi^\delta}{\mathrm{d}s} \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}.$$

In Eq. (93), we have used Eqs. (2) and (5) to rewrite the derivative of $\boldsymbol{\phi}(\boldsymbol{\xi}(s))$.

- **Update D.** Solve for $\boldsymbol{u}_\mathrm{D}^n$ and $\dot{\boldsymbol{u}}_\mathrm{D}^n$ the BVPs given by Eqs. (74) and (77) with BCs (75), (78) and

$$\boldsymbol{u}_\mathrm{D}^n=\boldsymbol{u}_\mathrm{E}^n \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}, \tag{94}$$

$$\dot{\boldsymbol{u}}_\mathrm{D}^n=\dot{\boldsymbol{u}}_\mathrm{E}^n \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}. \tag{95}$$

- **Update F.** Solve the BVPs given by (60) and (70) and BCs (61) to (63), (65), (71) and (73) and

$$\overline{\boldsymbol{v}}_\mathrm{F}=\dot{\boldsymbol{u}}_\mathrm{E}^n \text{ on } \partial\Omega^\mathrm{r}_{\bigcirc}, \tag{96}$$

which replaces Eq. (65) for a rigid body, and solve Eq. (68).

An example of the coupled dynamics of F and E is shown in Fig. 5, where we show the interaction of a F with a hydrogel (E). Here, $\partial\Omega^{\mathrm{r}}_{\circ}$ is an ellipse with semi-axes $a = 0.2\,\mathrm{m}$, $b = 0.1\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}} = (0.4\,\mathrm{m}, 0.2\,\mathrm{m})$, and $\partial\Omega^{\mathrm{r}}_{\odot}$ has radius $r = 0.0125\,\mathrm{m}$ and center located at $\mathbf{x}_{\mathrm{r}}$. The channel dimensions and F parameters have been taken from the FEAT2D DFG 2D-3 benchmark example for a fluid flow around a cylinder [16, 17]. The E density is $\rho_{\mathrm{E}} = 10^3\,\mathrm{Kg/m}^2$ [21]. The storage modulus, which approximates the shear modulus μ [22] is $\sim 10\,\mathrm{Kg/\,sec}^2$ [23]. Given a E Poisson ratio [3] of $\nu \sim 0.49$ [24], we obtain $K \sim 5 \times 10^2\,\mathrm{Kg/\,sec}^2$ from the relation $K = 2\mu(1+\nu)/[3(2\nu - 1)]$ [3]. The F inflow velocity profile is a Poiseuille flow, Eq. (80), with $v_0 = 1\,\mathrm{m/\,sec}$. Finally, the D mesh stiffness exponent has been chosen to be $\zeta = 3$ [2].

III. FLUID AND HELFRICH MEMBRANE

In this Section, we will build on the analysis of Section II, and complexify the structure of the elastic body with which F interacts by considering, instead of an elastic material, an Helfrich membrane (M) [1, 25].

For this problem, we will consider the geometry depicted in Fig. 6. A vertical channel with mirror symmetry about its mid axis is considered. The left boundary of the channel is $\partial\Omega_{\square}$, and the boundary $\partial\Omega_{\square}$ lies on the symmetry axis. Fluid is injected from bottom to top on $\partial\Omega_{\square}$. M lies on the top boundary of F, $\partial\Omega_{\square}$, which coincides with the M manifold $\partial\Omega_{\_\_}$. Finally, we denote by Ω the region enclosed by the boundaries, and all the quantities above will be considered in the r and c configurations.

The geometry of Fig. 6 may represent F being confined in a large reservoir, lying on bottom of the channel, where the channel is a vertical neck of the reservoir, covered on top by M.

A. Helfrich membrane motion

Both the physical nature and mathematical description of M is significantly more complex than that of E. First, unlike E, M has a fluid structure: its material elements can flow both tangentially and normally to it. Such fluid behavior requires an Eulerian description, which then needs to be connected to the Lagrangian description used to describe D. Second, M is described by fourth-order PDEs [26], which are significantly more complex than the second-order PDEs which describe E [12]. Third, the geometrical description of M requires fixing an appropriate *gauge*. In fact, a general parametrization of M is given by two functions $\mathbf{x}^{\mathrm{M}}_{\mathrm{c}}(x^1)$. Both of these functions depend on the curvilinear coordinate x^1 which is defined in the r domain of M, $\partial\Omega^{\mathrm{r}}_{\_\_}$, see Fig. 6—note that here x^i differs from the Cartesian coordinate x^α. Despite this pair of functions, a single PDE determines the M shape [26]. To understand this redundancy, consider the stretching field $\nu > 0$, defined by

$$\nu^2 \equiv \boldsymbol{e}_1 \cdot \boldsymbol{e}_1, \tag{97}$$

$$\boldsymbol{e}_i \equiv \frac{\partial \mathbf{x}^{\mathrm{M}}_{\mathrm{c}}}{\partial x^i}. \tag{98}$$

It is then easy to realize that two parametrizations with different νs may yield the same physical shape. Such redundancy may be removed by a gauge fixing, i.e., by adding a constraint on $\mathbf{x}^{\mathrm{M}}_{\mathrm{c}}$, or its derivatives. Two common gauge choices are the Monge [27, 28], or the arc-length parametrization [1], in which one sets

$$\nu = 1. \tag{99}$$

The arc-length gauge has been used to describe, for instance, the steady state for membrane tubules [1]. However, while a static constraint such as (99) is suited to static shapes, it cannot describe the M dynamics, such as the one of M interacting with F. In fact, as M is deformed, it will also stretch: as a result, Eq. (99) must be replaced by a time-dependent gauge—see below.

We describe M with the Lagrangian approach, along the lines of the description of D in Section I and E in Section II. The M displacement field is

$$\boldsymbol{u}_{\mathrm{M}}(x^1) \equiv \mathbf{x}^{\mathrm{M}}_{\mathrm{c}}(x^1) - \mathbf{x}^{\mathrm{M}}_{\mathrm{r}}(x^1), \tag{100}$$

where we choose the r configuration as a straight line

$$\mathbf{x}^{\mathrm{M}}_{\mathrm{r}}(x^1) \equiv (x^1, h),\ 0 \leq x^1 \leq L, \tag{101}$$

which is shown in Fig. 6A. In addition, we introduce the angle ψ formed by the tangent to M with the x^1 axis [1], defined by

$$e^1_1 = \nu \cos\psi, \tag{102}$$

$$e^2_1 = -\nu \sin\psi. \tag{103}$$

Finally, the M metric tensor is defined by [7]

$$g_{ij} \equiv \boldsymbol{e}_i \cdot \boldsymbol{e}_j. \tag{104}$$

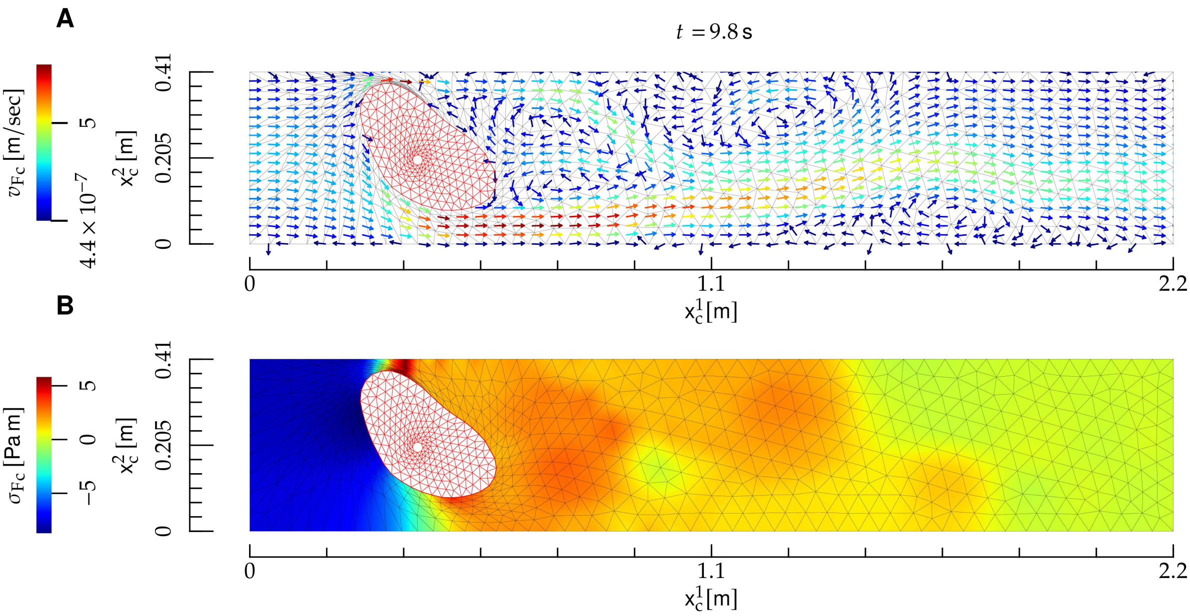

FIG. 5: Interaction between a bulk fluid and an elastic body (E) (a hydrogel). The notation is the same as in Fig. 2, and the elastic body is depicted as a red mesh. Here, the Reynolds number is $R \sim 2\times 10^2$.

In what follows, latin indices $i, j, k, \cdots$ will be used to denote vectors and tensors related to the M manifold; their position as upper and lower indexes is thus important [7]. Because the differential manifold of M is one dimensional, indexes $i, j, k, \cdots$ may take only one value, i.e., $i = j = k = \cdots = 1$. However, in formal relations such as Eq. (104), we will use the general notation $i, j, k, \cdots$ for the sake of generality, and for potential extensions of such relations to two-dimensional manifolds [28].

In order to work out the dynamical equations for M, let us derive the force exerted by F on M. The force exerted on a line element dl_c of M is [5]

$$f_{\mathrm{F}\,\alpha}(\sigma_{\mathrm{Fc}}, \nu, \psi) = -\sigma^{\mathrm{c}}_{\mathrm{F}\,\alpha\beta}\hat{n}^{\mathrm{c}\,\gamma}(\nu, \psi), \tag{105}$$

where the F stress tensor is evaluated at the c coordinate $\mathbf{x}_c^{\mathrm{M}} \in \partial\Omega^{\mathrm{c}}_{\square}$ corresponding to x^1. The components of the force (105) tangential and normal to M are [28]

$$f^i_{\mathrm{F}\,t}(\sigma_{\mathrm{Fc}}, \nu, \psi) = g^{ij}(\nu) e^{\alpha}_j(\nu, \psi) f_{\alpha}(\sigma_{\mathrm{Fc}}, \nu, \psi) \tag{106}$$

$$f_{\mathrm{F}\,n}(\sigma_{\mathrm{Fc}}, \nu, \psi) = \hat{n}^{\mathrm{c}\,\alpha}(\nu, \psi) f_{\alpha}(\sigma_{\mathrm{Fc}}, \nu, \psi), \tag{107}$$

where in Eq. (106) we used the feature that the metric tensor depends on ν only, see Eqs. (97) and (104).

The equations of motion for M are [12, 29, 30]

$$\nabla_i v^i_{\mathrm{M}} - 2H w_{\mathrm{M}} = 0, \tag{108}$$

$$\rho_{\mathrm{M}}\left(\frac{\partial v^i_{\mathrm{M}}}{\partial t} + v^j_{\mathrm{M}}\nabla_j v^i_{\mathrm{M}} - 2 v^j_{\mathrm{M}} w_{\mathrm{M}}\, b^i_j - w_{\mathrm{M}}\nabla^i w_{\mathrm{M}}\right) = \nabla^i \sigma_{\mathrm{M}} + f^i_{\eta_{\mathrm{M}}\,t} + f^i_{\mathrm{F}\,t}, \tag{109}$$

$$\rho_{\mathrm{M}}\left[\frac{\partial w_{\mathrm{M}}}{\partial t} + v^i_{\mathrm{M}}(v^j_{\mathrm{M}} b_{ji} + \nabla_i w_{\mathrm{M}})\right] = 2\sigma_{\mathrm{M}} H + f_{\kappa} + f_{\eta_{\mathrm{M}}\,n} + f_{\mathrm{F}\,n}, \tag{110}$$

$$\frac{\partial \boldsymbol{u}_{\mathrm{M}}}{\partial t} = w_{\mathrm{M}}\hat{n}^{\mathrm{c}}, \tag{111}$$

$$\frac{\partial u^1_{\mathrm{M}}}{\partial x^1} = -1 + \nu\cos\psi, \tag{112}$$

and they hold for $x^1 \in \partial\Omega^{\mathrm{r}}_{-}$. Here, v^i_{M} and w are the M tangential and normal velocity, respectively, ∇ is the covariant derivative related to the Levi-Civita connection associated to g, ∇_{LB} the Laplace-Beltrami operator, b the second fundamental form, and H and K the mean and Gaussian curvature, respectively. We refer to [7, 12, 28, 30] for details on the differential-geometry framework, and we will now discuss each line in Eqs. (108) to (113). Equation (108) enforces mass conservation [5]. Equations (109) and (110) are the tangential and normal components of the NS equations, respectively. The first term in the RHS of Eq. (109) represnts the force on membrane elements induced by gradients of line tension, while the first term in the RHS of Eq. (110) constitutes Laplace pressure [31]. The other terms read

$$f_\kappa(\nu, \psi) \equiv \tag{114}$$
$$-2\kappa[\nabla_i \nabla^i H + 2H(H^2 - K)],$$
$$f^i_{\eta_{\mathrm{M}}\, t}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv \tag{115}$$
$$\eta_{\mathrm{M}}[-\nabla_{\mathrm{LB}} v^i_{\mathrm{M}} -$$
$$2\left(b^{ij} - 2\, H\, g^{ij}\right) \nabla_j w_{\mathrm{M}} + 2K v^i_{\mathrm{M}}],$$
$$f_{\eta_{\mathrm{M}}\, n}(v_{\mathrm{M}}, w_{\mathrm{M}}, \nu, \psi) \equiv \tag{116}$$
$$2\eta_{\mathrm{M}}[(\nabla^i v^j_{\mathrm{M}}) b_{ij} - 2w_{\mathrm{M}}(2H^2 - K)].$$

In Eq. (114), f_κ is represents the resistance of M to bending, and it is derived from Helfrich model [25], while $f_{\eta_{\mathrm{M}}\, t}$ and $f_{\eta_{\mathrm{M}}\, n}$ in Eqs. (115) and (116) are, respectively, the tangential and normal component of the viscous force resulting from in-membrane flows [30]. Equation (111) is the time-evolution equation for the M shape, which evolves according to the normal $\hat{n}^c$, and it has been obtained from Eqs. (100) and (101). Equations (112) and (113) re-express the definitions of ν and ψ, and they result from Eqs. (98) and (100) to (103). Finally, ρ_{M}, σ_{M} and κ are, respectively, the density, line tension and bending rigidity of M.

We choose the following BCs for Eqs. (108) to (113):

$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{117}$$
$$v^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{118}$$
$$w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{119}$$
$$\nabla_i w_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{120}$$
$$\sigma_{\mathrm{M}} = \sigma_{\mathrm{M}\,\bullet\!-} \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{121}$$
$$\boldsymbol{u}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{122}$$

$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = -\,\nu \sin\psi, \tag{113}$$

$$u^1_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{123}$$
$$\frac{\partial u^2_{\mathrm{M}}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{124}$$

Equations (117), (119) and (122) enforce the condition that the left extremity of M, $\partial\Omega_{\bullet\!-}$, is clamped on the channel wall $\partial\Omega_{|\square}$. Equations (118), (120), (123) and (124) enforce the mirror symmetry about $\partial\Omega_{\square|}$. Namely, Eq. (118) ensures that the v_{M} vector field vanishes on $\partial\Omega_{-\!\bullet}$, because a nonzero value of this field on $\partial\Omega_{\square|}$ would break the axial symmetry. The condition (123) follows the same lines. In addition, the symmetry requires that scalar fields and vertical components of vector fields in Ω have zero slope with respect to x^1 at $\partial\Omega_{-\!\bullet}$, see Eqs. (120) and (124). Finally, Eq. (121) fixes the M tension on $\partial\Omega_{\bullet\!-}$.

B. Bulk fluid domain motion

The equation of motion for D are (33) and (41), with BCs

$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{125}$$
$$u^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{126}$$
$$\frac{\partial u^2_{\mathrm{D}}}{\partial \mathsf{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{127}$$
$$\boldsymbol{u}_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}, \tag{128}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{129}$$
$$\dot{u}^1_{\mathrm{D}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{130}$$
$$\frac{\partial \dot{u}^2_{\mathrm{D}}}{\partial \mathsf{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{131}$$
$$\dot{\boldsymbol{u}}_{\mathrm{D}} = \dot{\boldsymbol{u}}_{\mathrm{M}} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}. \tag{132}$$

Equation (125) ensures that D is not deformed on $\partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}$, cf. Eq. (39). Equations (126) and (127) follow from axial symmetry, and Eq. (128) from the requirement that D and M deformations stay conforming [2]. Finally, the BCs for $\dot{\boldsymbol{u}}_{\mathrm{M}}$, Eqs. (129) to (132), are derived along the same lines as Eqs. (125) to (128).

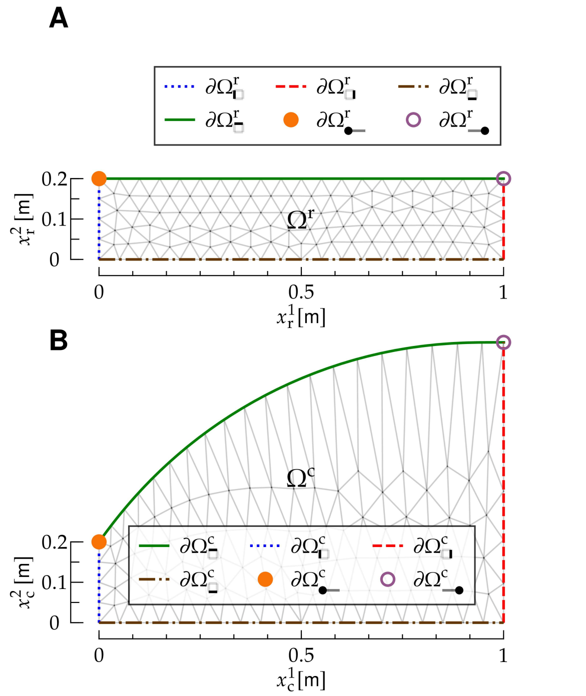

FIG. 6: Reference and current configuration for the interaction between a bulk fluid (F) and a Helfrich membrane (M). The figure follows the same notation as Fig. 1. The top edge of the F mesh, $\partial\Omega^{\mathrm{r}}_{\sqsupset}$, coincides with the one-dimensional manifold $\Omega^{\mathrm{r}}_{\_}$ of M.

C. Bulk fluid motion

The F equations of motion are Eqs. (13) and (14), with BCs

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \boldsymbol{v}_{\mathrm{F}\,\square} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{133}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}} = \mathbf{0} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{134}$$

$$v_{\mathrm{F}\,\mathrm{c}}^{1} = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{135}$$

$$\frac{\partial v_{\mathrm{F}\,\mathrm{c}}^{2}}{\partial \mathsf{x}^2_{\mathrm{c}}} = 0 \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{136}$$

$$\boldsymbol{v}_{\mathrm{F}\,\mathrm{c}}(\mathsf{x}_{\mathrm{c}}) = \left.\frac{\mathrm{d}\boldsymbol{u}_{\mathrm{M}}(x^1)}{\mathrm{d}t}\right|_{x^1=[\boldsymbol{\psi}^t(\mathsf{x}_{\mathrm{c}})]^1} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}, \tag{137}$$

$$\sigma_{\mathrm{F}\,\mathrm{c}} = \sigma_{\mathrm{F}\,\square} \text{ on } \partial\Omega^{\mathrm{c}}_{\square}. \tag{138}$$

According to Eq. (133), F is injected from the bottom edge of Ω^{c}, while Eq. (134) is a no-slip BC on the left wall $\partial\Omega^{\mathrm{c}}_{\square}$, and Eqs. (135) and (136) result from symmetry. Equation (137) results from the conforming motion [2] of F and M. Finally, Eq. (138) stems from the presence of the reservoir, in which the F tension is equal to $\sigma_{\mathrm{F}\,\square}$: At the boundary $\partial\Omega^{\mathrm{c}}_{\square}$ between the channel and the reservoir, the F tension matches the reservoir's.

The dynamics of the system is obtained as follows; the derivation is obtained along the lines of Sections I and II, and details can be obtained by referring to those Sections.

Given the M fields $v^{n-1,i}_{\mathrm{M}}$, $w^{n-1,i}_{\mathrm{M}}$, ${\sigma_{\mathrm{M}}}^{n-3/2}$, $\boldsymbol{u}^{n-3/2}_{\mathrm{M}}$, $\nu^{n-3/2}$, $\psi^{n-3/2}$, the D fields $\boldsymbol{u}^{n-1}_{\mathrm{D}}$, $\dot{\boldsymbol{u}}^{n-1}_{\mathrm{D}}$, and the F fields ${\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}}^{n-1}$, ${\sigma_{\mathrm{F}\,\mathrm{r}}}^{n-3/2}$, we determine the fields at the next time step as follows:

- **Update M.** Solve the BVP

$$\begin{aligned}\rho_{\mathrm{M}}\Big[\frac{\overline{v}^i_{\mathrm{M}} - v^{n-1,i}_{\mathrm{M}}}{\Delta t} + \left(\frac{3}{2}v^{n-1,j}_{\mathrm{M}} - \frac{1}{2}v^{n-2,j}_{\mathrm{M}}\right)\nabla^{n-1/2}_j V^i_{\mathrm{M}} - 2V^j_{\mathrm{M}}W_{\mathrm{M}}b^{n-1/2,i}_{\ \ j} - \\ W_{\mathrm{M}}\nabla^{n-1/2,i}W_{\mathrm{M}}\Big] = \\ \nabla^{n-1/2,i}{\sigma_{\mathrm{M}}}^* + f^i_{\eta\mathrm{M}}\left(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2}\right) + \\ f^i_{\mathrm{F}\,t}(\boldsymbol{\sigma}_{\mathrm{F}\,\mathrm{r}}({\boldsymbol{v}_{\mathrm{F}\,\mathrm{r}}}^{n-1}, {\sigma_{\mathrm{F}\,\mathrm{r}}}^{n-3/2}, \boldsymbol{u}^{n-1}_{\mathrm{D}}), \nu^{n-1/2}, \psi^{n-1/2}),\end{aligned} \tag{139}$$

$$\rho_{\mathrm{M}}\left[\frac{\overline{w}_{\mathrm{M}} - w^{n-1}_{\mathrm{M}}}{\Delta t} + V^i_{\mathrm{M}}V^j_{\mathrm{M}}b^{n-1/2,j}_{\ \ i} + \left(\frac{3}{2}v^{n-1,i}_{\mathrm{M}} - \frac{1}{2}v^{n-3/2,i}_{\mathrm{M}}\right)\nabla^{n-1/2}_i W_{\mathrm{M}}\right] = \tag{140}$$

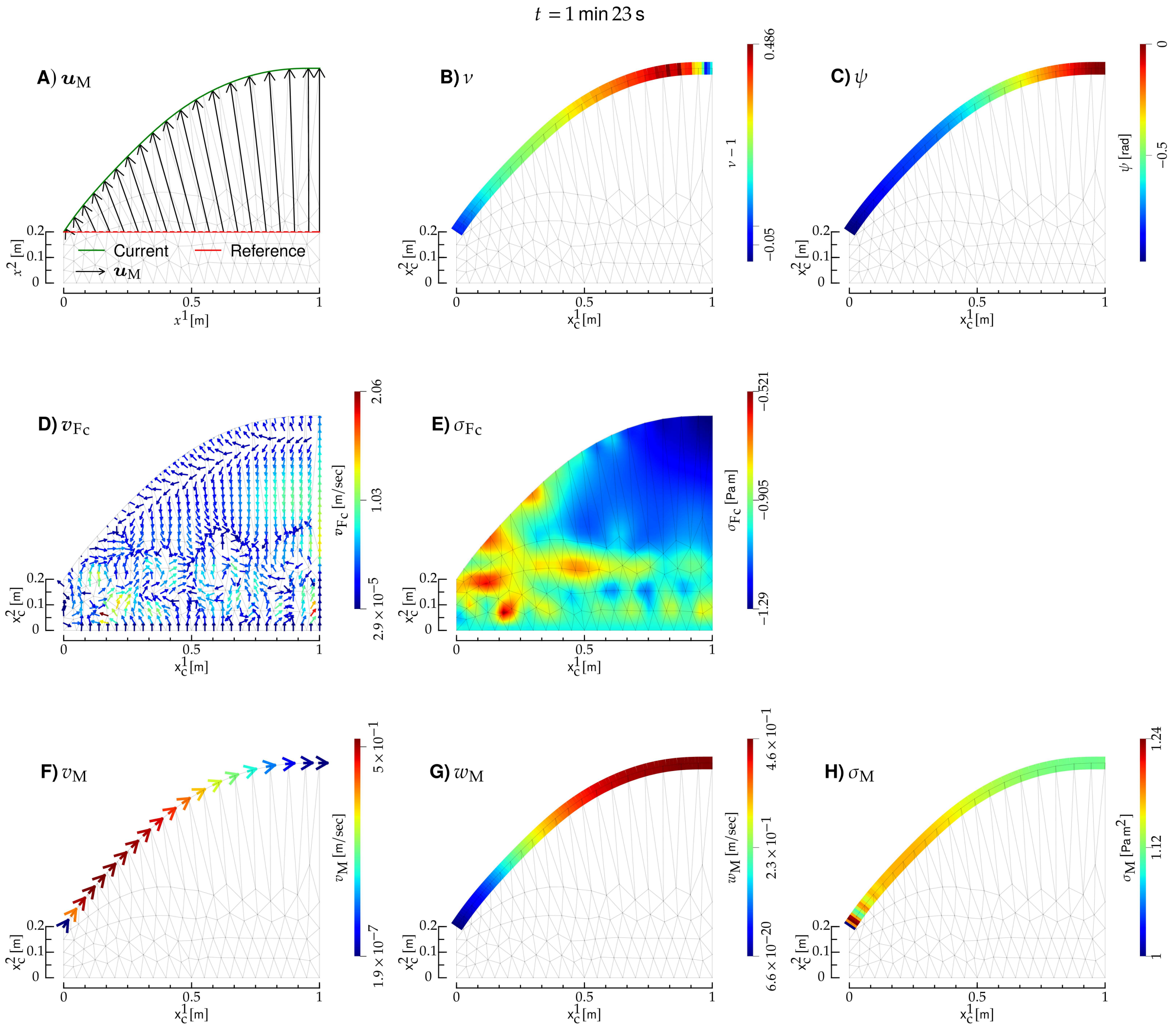

FIG. 7: Interaction between a bulk fluid (F) and a Helfrich membrane (M). Here, F is air, and M an elastomer, and the F Reynolds number is $R \sim 200$. **A**) Reference (r) and current (c) configuration of M (red and green curve, respectively), and M displacement field $\boldsymbol{u}_{\mathrm{M}}$ (black arrows). **B**) Stretch field of M. **C**) Tangent angle of M. **D**) and **E**): F velocity and tension, respectively; the notation is the same as in Fig. 2. **F**), **G**) and **H**): Tangential velocity, normal velocity and tension of M, respectively. All panels refer to the same instant of time, shown on top, and they display also the deformed mesh (gray lines).

$$f_\kappa(\nu^{n-1/2}, \psi^{n-1/2}) + 2\sigma_{\mathrm{M}}{}^* H^{n-1/2} + f_{\eta_{\mathrm{M}}\, n}(\boldsymbol{V}_{\mathrm{M}}, W_{\mathrm{M}}, \nu^{n-1/2}, \psi^{n-1/2}) + \\ f_{\mathrm{F}\, n}(\boldsymbol{\sigma}_{\mathrm{F\,r}}(\boldsymbol{v}_{\mathrm{F\,r}}{}^{n-1}, \sigma_{\mathrm{F\,r}}{}^{n-3/2}, \boldsymbol{u}_{\mathrm{D}}^{n-1}), \nu^{n-1/2}, \psi^{n-1/2}),$$

$$\nabla_i^{n-1/2}\nabla^{n-1/2,\, i}\phi_{\mathrm{M}} = \frac{\rho_{\mathrm{M}}}{\Delta t}(\nabla_i^{n-1/2}\overline{v}_{\mathrm{M}}^i - 2H^{n-1/2}\overline{w}_{\mathrm{M}}), \tag{141}$$

$$\rho_{\mathrm{M}}\frac{\overline{v}_{\mathrm{M}}^i - v_{\mathrm{M}}^{n,\, i}}{\Delta t} = -\nabla^{n-1/2,\, i}\phi_{\mathrm{M}}, \tag{142}$$

$$w_{\mathrm{M}}^n = \overline{w}_{\mathrm{M}}, \tag{143}$$

$$\frac{u_{\mathrm{M}}^{n-1/2,\, \alpha} - u_{\mathrm{M}}^{n-3/2,\, \alpha}}{\Delta t} = w_{\mathrm{M}}^{n-1}\hat{n}^{\mathrm{c}\, n-1/2,\, \alpha}, \tag{144}$$

$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,1}}{\partial x^1} = -\,1 + \nu^{n-1/2}\cos\psi^{n-1/2}, \tag{145}$$

$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1} = -\,\nu^{n-1/2}\sin\psi^{n-1/2}, \tag{146}$$

with BCs

$$\overline{v}^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{147}$$
$$\overline{v}^i_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{148}$$
$$\overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{149}$$
$$\nabla_i^{n-1/2}\overline{w}_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{150}$$
$$\phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{151}$$
$${n_{\mathrm{M}}}_i^{n-1/2}\nabla^{n-1/2,\,i}\phi_{\mathrm{M}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-}, \tag{152}$$
$$\boldsymbol{u}_{\mathrm{M}}^{n-1/2} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\bullet\!-}, \tag{153}$$
$$u_{\mathrm{M}}^{n-1/2,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}, \tag{154}$$
$$\frac{\partial u_{\mathrm{M}}^{n-1/2,\,2}}{\partial x^1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{-\!\bullet}. \tag{155}$$

Here, Eqs. (139) and (140) are the PDEs for the approximate velocities, Eq. (141) for pressure correction, and Eqs. (142) and (143) for the velocity correction. Equation (144) is the time-evolution equation for the displacement field, and Eqs. (145) and (146) the discrete versions of Eqs. (112) and (113), respectively. In Eq. (59), n_{M} is the unit normal to $\partial\Omega_{-}$: It lies in the tangent bundle of Ω_{-}, is directed outside Ω_{-}, and is normalized according to ${n_{\mathrm{M}}}^i{n_{\mathrm{M}}}_i = 1$ [7, 12, 32].

This step determines the M fields $v_{\mathrm{M}}^{n,i}$, $w_{\mathrm{M}}^{n,\,i}$, ${\sigma_{\mathrm{M}}}^{n-1/2}$, $\boldsymbol{u}_{\mathrm{M}}^{n-1/2}$, $\nu^{n-1/2}$, $\psi^{n-1/2}$.

- **Update D.** We determine the D displacement field and its timer derivative, $\boldsymbol{u}^n_{\mathrm{D}}$ and $\dot{\boldsymbol{u}}^n_{\mathrm{D}}$, from the BVP (74) and (77), with BCs given by the discrete version of Eqs. (125) to (132):

$$\boldsymbol{u}^n_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{156}$$
$$u_{\mathrm{D}}^{n,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{157}$$
$$\frac{\partial u_{\mathrm{D}}^{n,\,2}}{\partial \mathsf{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{158}$$
$$\boldsymbol{u}^n_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}}^{n-1/2} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}, \tag{159}$$
$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{160}$$
$$\dot{u}_{\mathrm{D}}^{n,\,1} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{161}$$
$$\frac{\partial \dot{u}_{\mathrm{D}}^{n,\,2}}{\partial \mathsf{x}^1_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{162}$$
$$\dot{\boldsymbol{u}}^n_{\mathrm{D}} = \boldsymbol{u}_{\mathrm{M}}^{n-1/2} \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}. \tag{163}$$

- **Update F.** We determine the F fields ${v_{\mathrm{F}}}_{\mathrm{r}}^n$ and ${\sigma_{\mathrm{F}}}_{\mathrm{r}}^{n-1/2}$ by solving the BVP (60) and (70) with BCs

$$\overline{\boldsymbol{v}}_{\mathrm{F}} = \boldsymbol{v}_{\mathrm{F}\,\underline{\square}} \text{ on } \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{164}$$
$$\overline{\boldsymbol{v}}_{\mathrm{F}} = \boldsymbol{0} \text{ on } \partial\Omega^{\mathrm{r}}_{|\square}, \tag{165}$$
$$\overline{v}^1_{\mathrm{F}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{166}$$
$$G(\boldsymbol{u}_{\mathrm{D}}^{n-1})_{\alpha 1}\frac{\partial V^2_{\mathrm{F}}}{\partial \mathsf{x}^\alpha_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{167}$$

$$\overline{\boldsymbol{v}}_{\mathrm{F}}(x^1, h) = \dot{u}_{\mathrm{M}}^{n-1/2}(x^1) \text{ on } \partial\Omega^{\mathrm{r}}_{\overline{\square}}, \tag{168}$$
$$\phi_{\mathrm{F}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\underline{\square}}, \tag{169}$$
$$\hat{n}^{\mathrm{r}\,\gamma}G_{\gamma\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})G_{\beta\alpha}(\boldsymbol{u}_{\mathrm{D}}^{n-1})\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}^\beta_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{|\square} \cup \partial\Omega^{\mathrm{r}}_{\overline{\square}}, \tag{170}$$
$$G(\boldsymbol{u}_{\mathrm{D}}^{n-1})_{\alpha 1}\frac{\partial\phi_{\mathrm{F}}}{\partial \mathsf{x}^\alpha_{\mathrm{r}}} = 0 \text{ on } \partial\Omega^{\mathrm{r}}_{\square|}, \tag{171}$$

and by solving Eq. (68).

An example of this dynamics is shown in Fig. 7, where we show the interaction between air (F) and an elastomer (M).

The channel has dimensions $L = 1\,\mathrm{m}$, $h = 0.2\,\mathrm{m}$. Proceeding along the lines of Section I, M parameters have been estimated by considering the parameters for a three-dimensional material, multiplying them by a unit length corresponding to the out-of-plane depth, and by the M thickness, chosen to be $\delta = 10^{-3}\,\mathrm{m}$. The M bending rigidity κ has been estimated from the relation $\kappa = E\delta^3/[12(1-\nu^2)]$, which expresses κ as a function of the Young modulus E, the Poisson ratio ν, and δ [3]. Typical values for an elastomer are $E \sim 10^7\,\mathrm{Pa}$ [33] and $\nu \sim 0.5$ [34], yielding $\kappa = 10^{-3}\,\mathrm{Kg\,m^3/\,sec^2}$. We chose $\rho_{\mathrm{M}} = 1\,\mathrm{Kg/m}$ and $\eta_{\mathrm{M}} = 1\,\mathrm{Kg\,m/\,sec}$ as the density and viscosity of an elastomer, respectively [35, 36], and we take $\sigma_{\mathrm{M}\,\bullet\!-} = 1\,\mathrm{N}$. The F parameters have been estimated from the air parameters: $\rho_{\mathrm{F}} = 1\,\mathrm{Kg/m^2}$, $\eta_{\mathrm{F}} = 10^{-5}\,\mathrm{Kg/\,sec}$ [18]. Also, we take $\sigma_{\mathrm{F}\,\underline{\square}} = -1\,\mathrm{N/m^2}$, and $\boldsymbol{v}^1_{\mathrm{F}\,\underline{\square}} = 0$, $\boldsymbol{v}^2_{\mathrm{F}\,\underline{\square}}(\mathsf{x}_{\mathrm{c}}) = \frac{3}{2L^2}\mathsf{x}^1_{\mathrm{c}}(2L - \mathsf{x}^1_{\mathrm{c}})v_0$, where $v_0 = 10^{-2}\,\mathrm{m/\,sec}$, and $\boldsymbol{v}^2_{\mathrm{F}\,\underline{\square}}$ corresponds to a Poiseuille flow [5]. Finally, the D exponent is $\zeta = 3$ [2].

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